① There are three types of molten salt reactor:

① Salt cooled → traditional solid fuel, but coolant is a molten salt

② still salt fueled → fuel is a molten salt within cladding, often also uses a molten salt coolant

③ flowing salt fueled → fuel is a molten salt that flows through the loop and doubles as a coolant

The advantages of MSRs include:

- operate near atmospheric pressures
- operate at higher temperatures
- have a strong negative reactivity coefficient due to thermal expansion
- fission products readily dissolve in molten salt

② Fissile / fertile material → causes fission

solvent → serves to lower melting point and decrease viscosity

oxygen getter → remove oxygen to minimize corrosion

fission products → result of fission

③ The cover gas system serves to remove fission product gasses from the loop to maintain redox conditions and protect against corrosion.

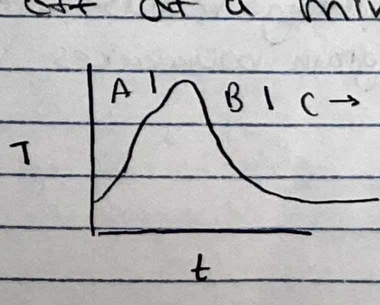
④ The UF_4/UF_3 ratio is representative of the redox potential and is normally maintained around 100. If the ratio is too high, it can cause corrosion of the metal cladding, but if it is too low, it can generate UC phases from graphite in the core. The ratio is constantly changing due to UF_4 fissions that generate an excess of F atoms with oxidizing potential, but it is important to maintain the correct ratio to prevent corrosion.

⑤ Corrosion of the cladding occurs through three primary processes. First, reaction with a metallic oxide layer that is generally created during fabrication. This process typically occurs immediately, resulting in rapid initial corrosion where the metallic oxides turn into metallic fluorides.
The second process is reactions with dissolved impurities in the molten salt. Of primary concern is oxygen, which is highly reactive.
Lastly includes reactions with the normal molten salt constituents themselves, including fissile material and fission products.
Overall, cladding corrosion will typically occur rapidly in the beginning due to the oxide layer, then the rate is determined by chromium diffusion to the boundary.

Exam 3

(6) Carbide / nitride fuels have a higher fissile density than oxide fuels and can generate more power. They also have a higher thermal conductivity and higher melting point than metallic fuels, which allows for a larger diameter fuel pin, thus fewer fuel pins, and a more compact and cheaper core. Additionally, nitride fuels can have longer fuel cycles and are compatible with PUREX reprocessing.

(8) The first stage of temperature evolution in He-bonded C/N fuels involves a rapid increase in temperature due to initial cracking of the fuel that causes separation and migration into the fuel-cladding gap. The second stage might begin with a small temperature rise as the cracks resinter and the bond gap increases momentarily, but quickly a turnaround occurs when unrestrained fuel swelling decreases the bond gap and increases thermal conductivity. The third stage begins when the gap is fully-closed, the temperature levels off at a minimum, and FCMI begins.



(9) Carbides restructure in four stages. Stage I occurs in the center of the fuel pin at maximum temperatures and involves a highly porous microstructure. Stage II is an intermediate stage that only occurs at high heat up rates and involves columnar grains / pores. Stage III involves grain growth and bubbles in grain boundaries. Stage IV exists in the coolest outer region and is the as-fabricated microstructure. Nitride restructuring is similar, but Stage II has not been observed.

(10) The primary FCCI concern in carbide fuels is carburization of the cladding. This process involves carbon migration towards the cladding, partially driven by a thermal gradient, and passage through the bond material to reach the cladding. In He-bonded fuel, carbon transfer through the bond occurs as CO. In Na-bonded fuel, carbon directly dissolves in sodium - a faster process that makes carburization worse in Na-bonded fuel pins. Carbon that reaches the cladding reacts with Chromium and precipitates out at grain boundaries, and then, crystallographic planes.

12 The C/M-ratio decreases with burnup. If the ratio is too small metallic actinide phases /metallic precipitants can form, but if the ratio is too high, it increases carburization of the cladding.

The N/M ratio increases with burnup, which can lead to nitrogenation of the cladding.

11 The primary cause of failure in UC/UN fuels is FCMI due to fuel swelling. These fuels have higher thermal conductivities, thus lower temperatures and less fission gas release, but higher swelling rates.

7 The two primary pin designs are He-bonded and Na-bonded. He-bonded pins have a lower thermal conductivity and higher fuel temperatures, and therefore, require a larger bond gap than the Na-bonded pins. The Na-bond is a larger corrosion concern and the sodium purity must be considered.